In groups of 2–3, answer the following questions. For any calculation-based question, you only need to set up the calculation by writing the appropriate formula and substituting the given values. You do not need to carry out the calculation.

A large grocery store chain is preparing a new marketing campaign and wants to better understand its customers. Through data analytics, the research team took a random sample of 40 regular customers and recorded:

- Average transaction amount: the average amount of money the customer spends each time they make a purchase.
- Payment method: whether the customer typically pays with cash, card, or check. A "success" is considered payment by card.

1. What is the population of interest?
2. What is the observational unit?
3. For each variable, identify whether the variable is categorical or quantitative, and write out the parameter in both words and symbols.

Researchers decided to first look at payment method. From the sample of 40 regular customers, they found that 29 of them (72.5%) preferred to pay with card and want to test whether this is less than the national average of 84%.

4. In symbols, write the hypothesis test that the researchers are conducting.
5. Set up the calculation for the standardized statistic.
6. Set up the calculation for the 95% confidence interval.
7. The p-value was found to be 0.02363. Using a significance level of 0.05, what conclusion would the researchers make?

When analyzing the average transaction amount per customer, the average amount spent was \$92 with a standard deviation of \$45. This distribution of average transaction amounts was found to be approximately symmetric without strong skew.

8. Set up the calculation for the 95% confidence interval for the mean amount spent per transaction.
9. The 95% confidence interval was found to be (\$77.77, \$106.23). Explain what this confidence interval means in context.